

AReal: A Large-Scale Asynchronous Reinforcement Learning System for *Agentic Workflows*

Open-source Project on: <https://github.com/inclusionAI/AReal>

Task Specification: Agentic Workflow

The hybrid of:

- Long-context reasoning
- Multi-tool interaction
- Multi-turn user interaction

```

• Read(file_path: "components/ChatArea.tsx", offset: 190, limit: 30)_
  | // Introducing an error by removing these required props
  | }
  | ... (+27 lines)

• Found the issue. Let's fix the type definition at the correct location:

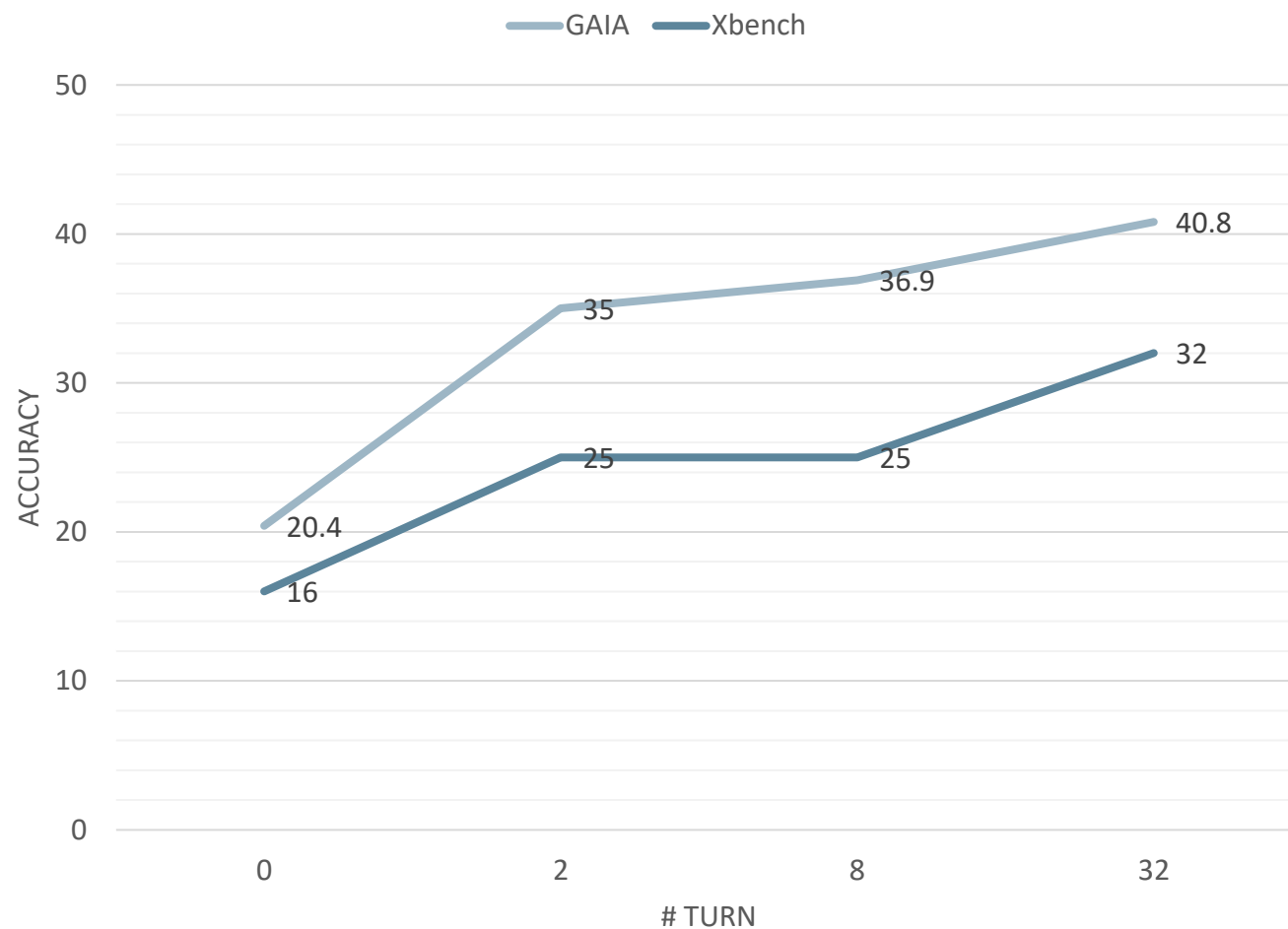
• Update(file_path: components/ChatArea.tsx)_
  | Updated components/ChatArea.tsx with 3 additions and 1 removal
  | 189   showAvatar: boolean;
  | 190   // Introducing an error by removing these required props
  | 190   selectedKnowledgeBase: string;
  | 191   setSelectedKnowledgeBase: (knowledgeBaseId: string) => void;
  | 192   knowledgeBases: KnowledgeBase[];
  | 193 }

• Herding_ (126s - esc to interrupt)

>
! for bash mode - / for commands - tab to undo      \= for newline

```

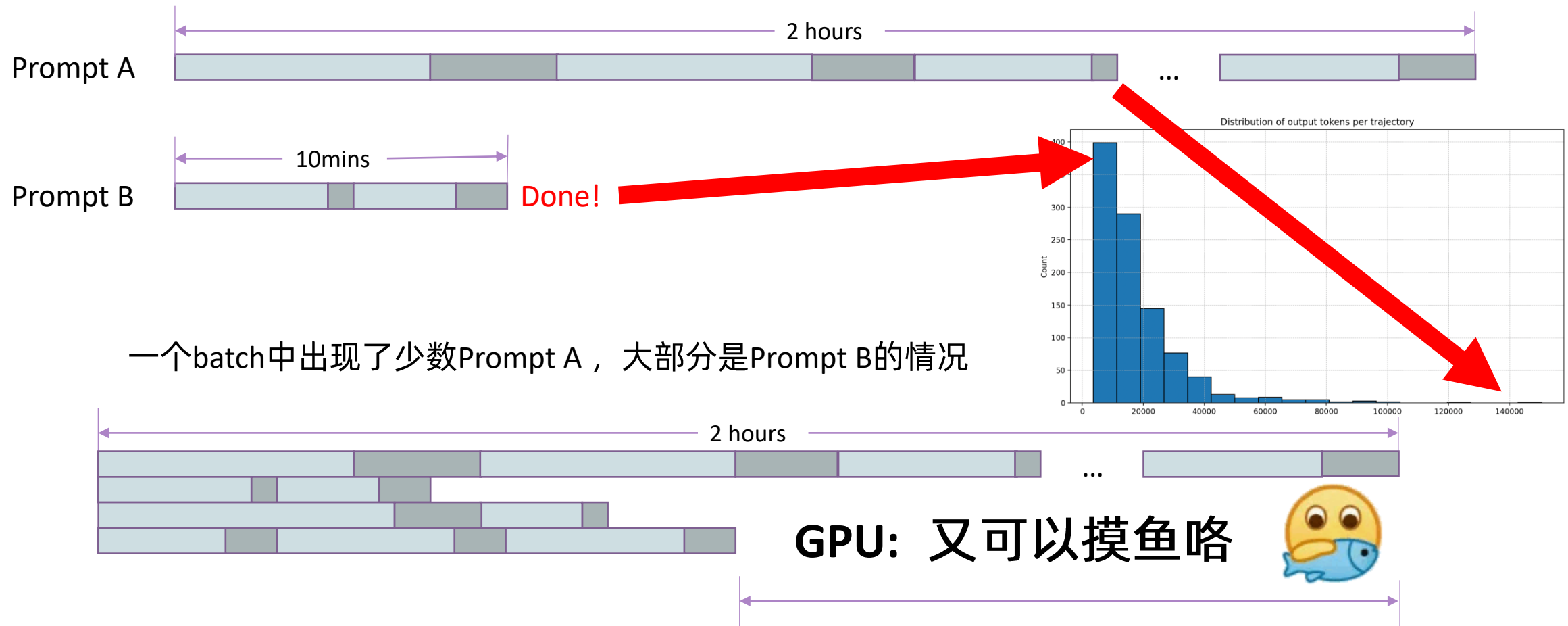
Search Agent交互轮数对结果的影响



测试时，更多的轮数能取得更好的结果

Problem 1: Inference Device Under-Utilization

RL 训练中放松对轮数的限制：8 $\rightarrow$ 128

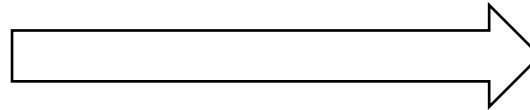


Problem 2: Hard to **scale up**

My GPU cluster



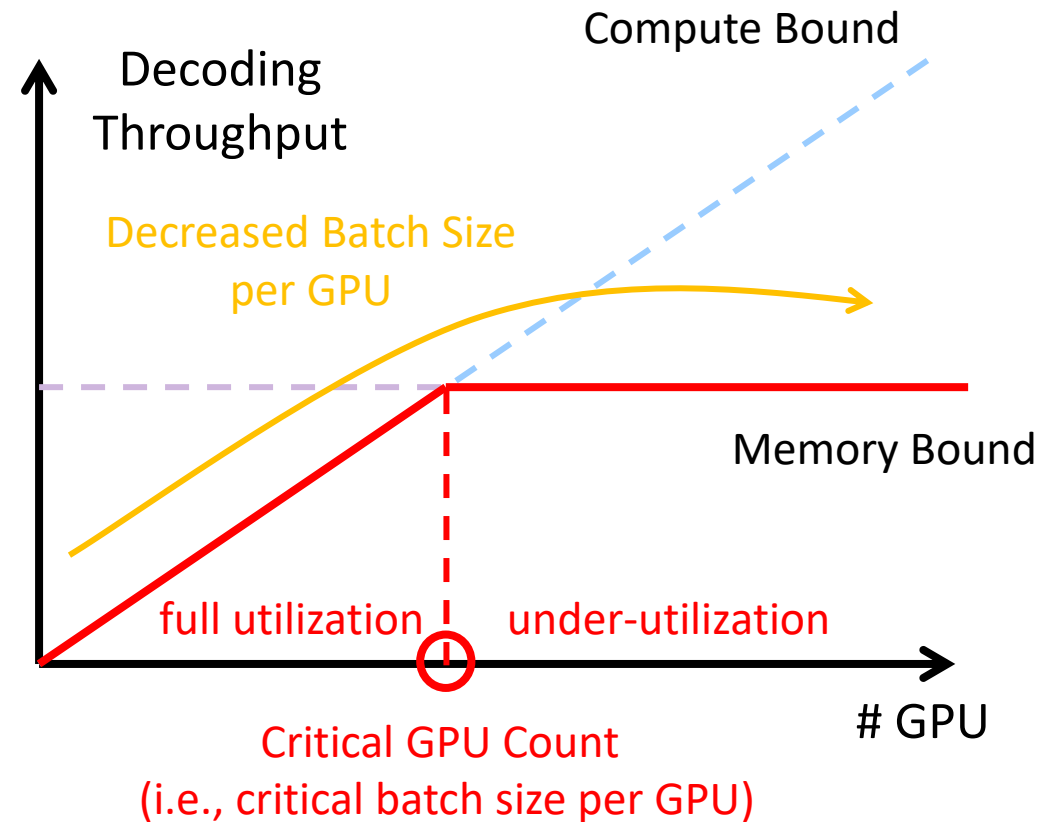
RL Experiment: 100 hours



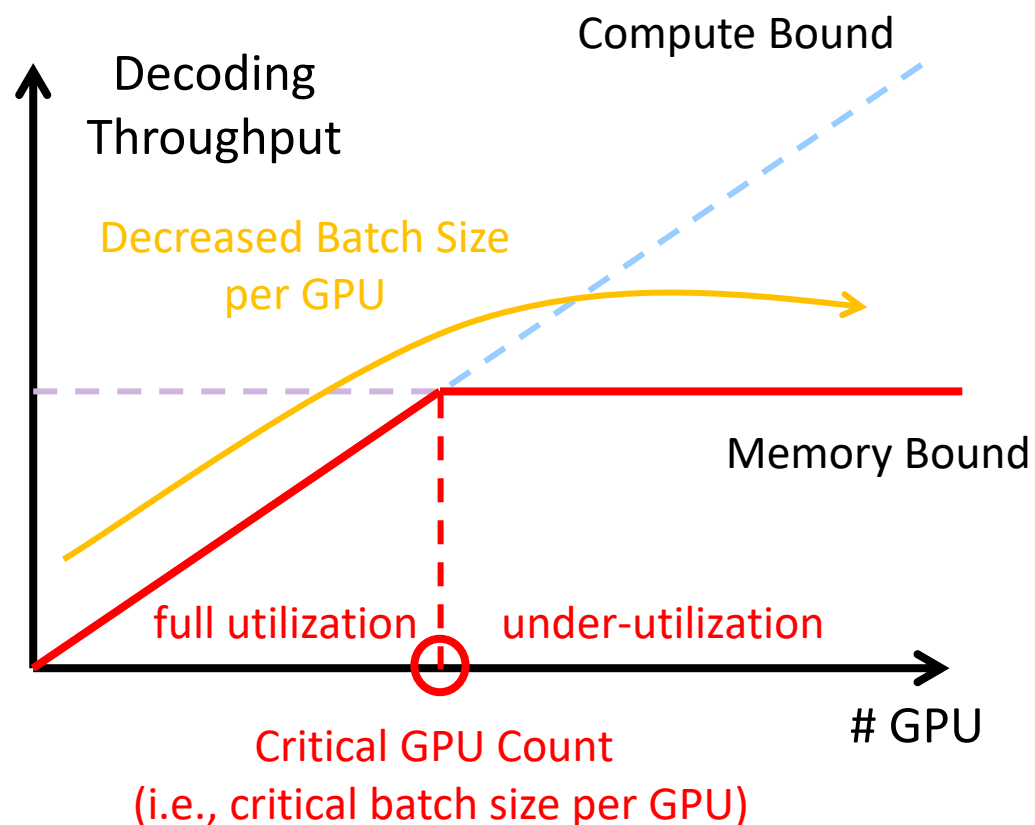
The same RL Experiment: 50 hours?

Problem 2: Hard to **scale up**

The change of decoding throughput with more GPUs



Problem 2: Hard to **scale up**



The critical batch size is determined by the hardware specification

- Definition: Budget = Compute (TFLOPS)-to-Bandwidth (TB/S) ratio
- Optimal number of tokens in a batch

Hardware	BF16 Compute	HBM Bandwidth	Budget
A100	312 TFLOPS	2.0TB/S	156
H100	1979 TFLOPS	3.4TB/s	582

Table 1: Specifications of NVIDIA A100 and H100 GPU and their corresponding budgets.

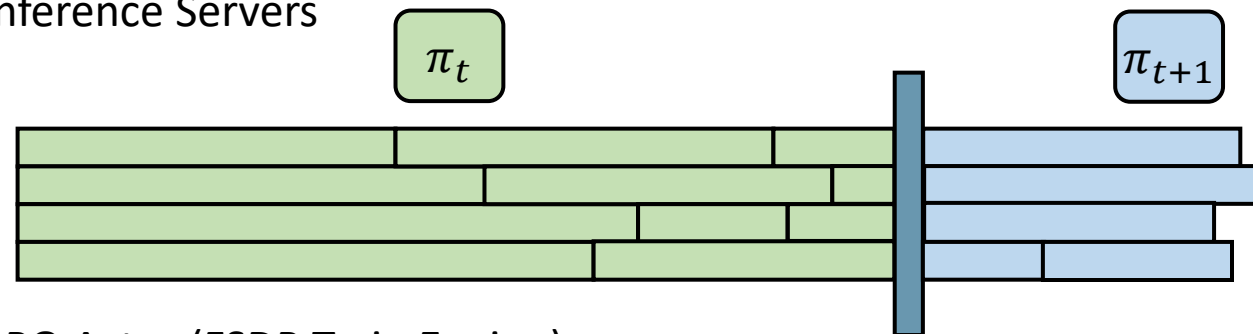
Reference:

1. Transformer Inference Arithmetic (<https://kipp.ly/transformer-inference-arithmetic/>).
2. SpecInfer: Accelerating Generative Large Language Model Serving with Tree-based Speculative Inference and Verification (ASPLOS 24)

Our Solution: **Fully** Asynchronous Training

训练的同时推理引擎满负载工作

Inference Servers



推理中断+参数更新



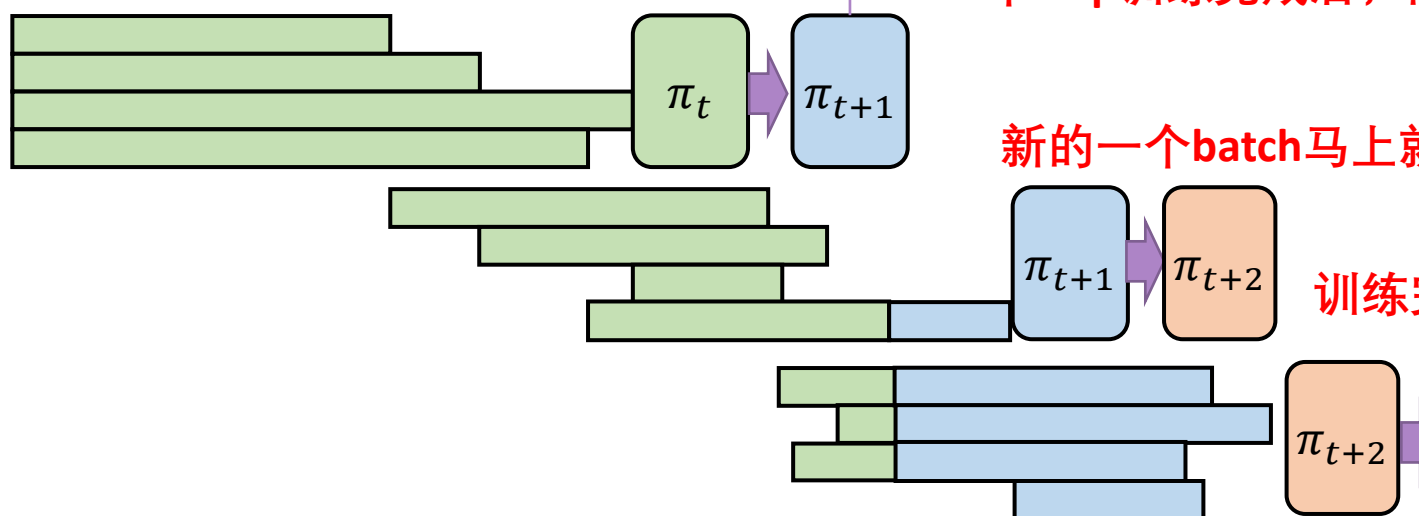
一条数据的rollout流程

PPO Actor (FSDP Train Engine)

一个step训练完成后，同步参数到推理引擎

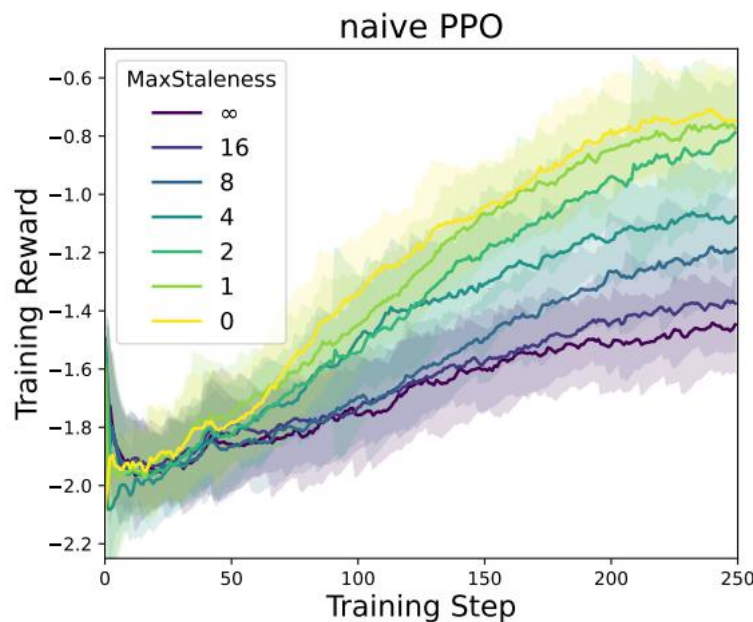
新的一个batch马上就准备好了，又可以开始训练

训练完成，新一轮数据无缝衔接

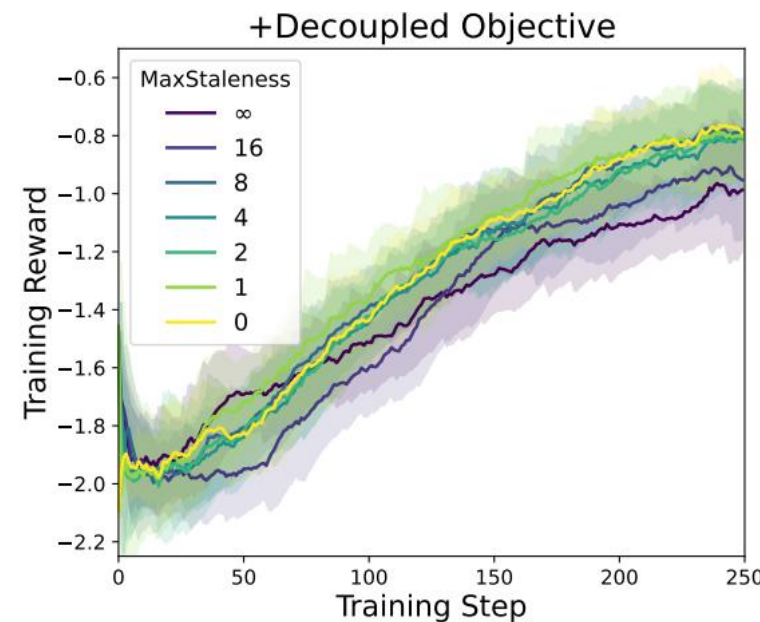


Algorithm-System Co-Design

- Explicit staleness/Off-policyness control
- Decoupled PPO Objective



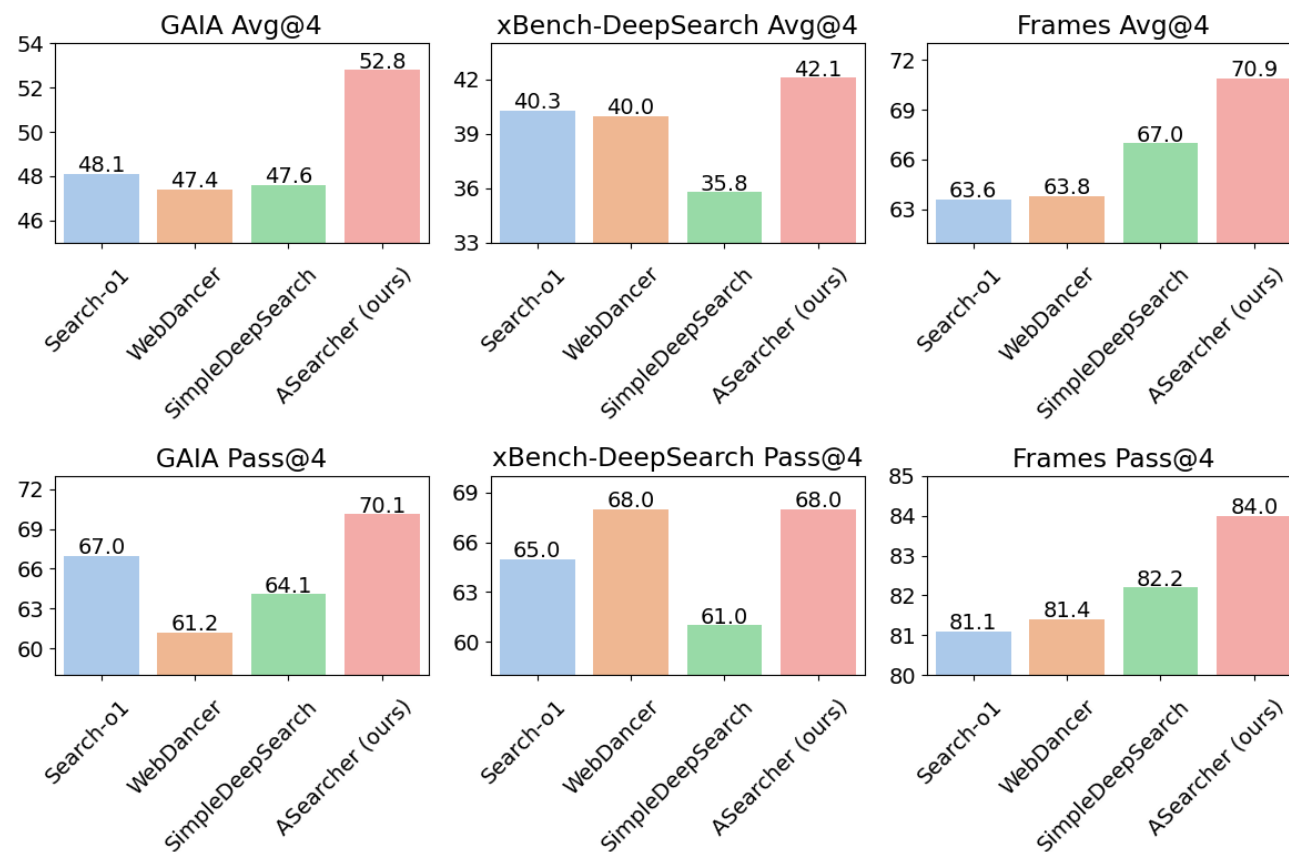
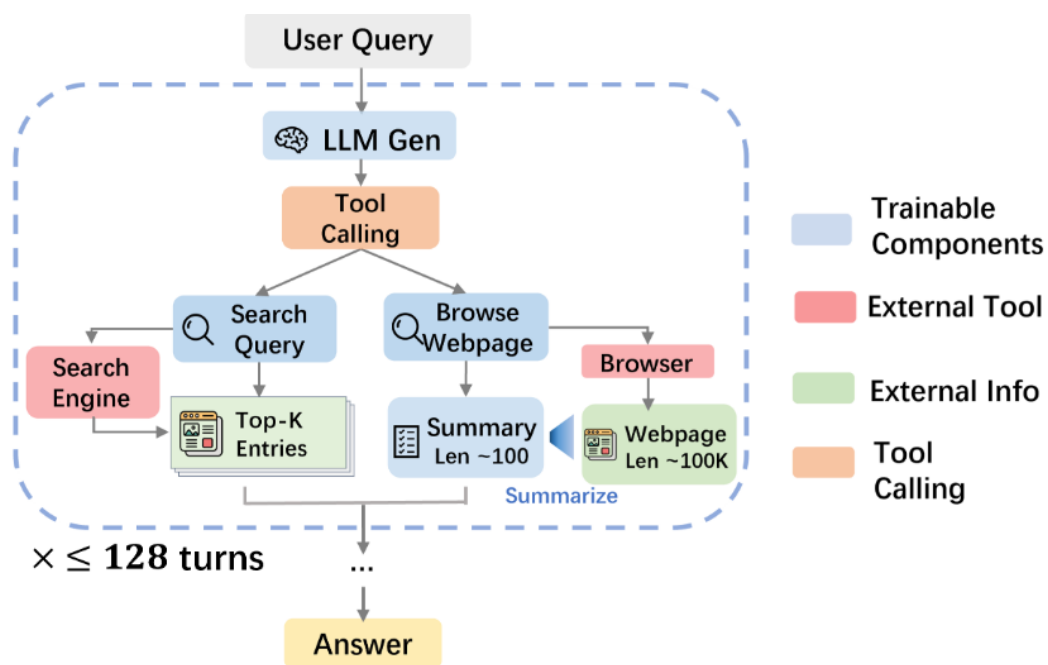
(a) Learning curves with naive PPO.



(b) Learning curves with eq. (5).

Project based on AReaL: ASearcher

~8000 GPU hours (H800) based on QwQ 32B



Algorithm-Driven API Design of AReal as an Agentic RL Framework

<https://github.com/inclusionAI/AReal>

What AReal code looks like

```
class RLVRWorkflow:

    async def arun_episode(self, engine: InferenceEngine, data):
        client = ArealOpenAI(engine, self.tokenizer)
        completions = await asyncio.gather(
            [
                client.chat.completions.create(
                    messages=data["messages"],
                    frequency_penalty=self.gconfig.frequency_penalty,
                    max_completion_tokens=self.gconfig.max_new_tokens,
                    stop=self.gconfig.stop,
                    store=True,
                    temperature=self.gconfig.temperature,
                    top_p=self.gconfig.top_p,
                )
                for _ in range(self.gconfig.n_samples)
            ]
        )
        rewards = await self.async_reward_fn(completions)
        for reward, c in zip(rewards, completions):
            client.set_reward(c.id, reward)
            # Log reward.
            stats_tracker.get(self.rollout_stat_scope).scalar(reward=reward)
        return client.export_completions()
```

What AReal code looks like

```
def main_grpo():
    config, _ = load_expr_config(sys.argv[1:], GRPOConfig)
    config: GRPOConfig
    rank = int(os.getenv("RANK"))
    world_size = int(os.getenv("WORLD_SIZE"))
    tokenizer = load_hf_tokenizer(config.tokenizer_path)
    # Create dataset and dataloaders
    train_dataloader = StatefulDataLoader(
        get_gsm8k_dataset("train", rank, world_size),
        batch_size=config.train_dataset.batch_size // world_size,
        shuffle=config.train_dataset.shuffle,
        num_workers=config.train_dataset.num_workers,
        collate_fn=lambda x: x,
        drop_last=config.train_dataset.drop_last,
    )
    # Initialize inference engine
    rollout = RemoteSGLangEngine(config.rollout)
    # Initialize train engine
    actor = FSDPPPOActor(config=config.actor)
    ref = None
    if config.actor.kl_ctl > 0 and config.ref is not None:
        ref = FSDPPPOActor(config=config.ref)
    workflow = RLVRWorkflow(
        reward_fn=gsm8k_reward_fn,
        gconfig=config.gconfig,
        tokenizer=tokenizer,
        enable_thinking=False,
    )
```

```
# Run training.
total_epochs = config.total_train_epochs
steps_per_epoch = len(train_dataloader)
max_steps = total_epochs * steps_per_epoch
data_generator = iter(train_dataloader)
for global_step in range(max_steps):
    epoch = global_step // steps_per_epoch
    step = global_step % steps_per_epoch
    try:
        data = next(data_generator)
    except StopIteration:
        data_generator = iter(train_dataloader)
        data = next(data_generator)
    batch = rollout.rollout_batch(data, workflow=workflow)
    batch = batch.to(actor.device)

    logp = actor.compute_logp(batch)
    batch["prox_logp"] = logp
    if ref is not None:
        batch["ref_logp"] = ref.compute_logp(batch)
    actor.compute_advantages(batch)
    stats = actor.ppo_update(batch)
    actor.step_lr_scheduler()
    print(epoch, step, global_step, stats)
```

Abstraction Level 1: Orchestration

- Loading datasets
 - Dataset processing
- Initializing models
 - Whether to use a reward/reference/critic model
- Training loop
 - Data loading logic
 - Hybrid multiple training algorithms (e.g., one-step SFT + one-step RL)
 - Whether to update weights and how
- Agent
 - Agentic workflow

Abstraction Level 1: Orchestration

```
def main_grpo():
    config, _ = load_expr_config(sys.argv[1:], GRPOConfig)
    config: GRPOConfig
    rank = int(os.getenv("RANK"))
    world_size = int(os.getenv("WORLD_SIZE"))
    tokenizer = load_hf_tokenizer(config.tokenizer_path)
    # Create dataset and dataloaders
    train_dataloader = StatefulDataLoader(
        get_gsm8k_dataset("train", rank, world_size),
        batch_size=config.train_dataset.batch_size // world_size,
        shuffle=config.train_dataset.shuffle,
        num_workers=config.train_dataset.num_workers,
        collate_fn=lambda x: x,
        drop_last=config.train_dataset.drop_last,
    )
    # Initialize inference engine
    rollout = RemoteSGLangEngine(config.rollout)
    # Initialize train engine
    actor = FSDPPPOActor(config=config.actor)
    ref = None
    if config.actor.kl_ctl > 0 and config.ref is not None:
        ref = FSDPPPOActor(config=config.ref)
    workflow = RLVRWorkflow(
        reward_fn=gsm8k_reward_fn,
        gconfig=config.gconfig,
        tokenizer=tokenizer,
        enable_thinking=False,
    )
```

Trainer-Free Implementation

```
# Run training.
total_epochs = config.total_train_epochs
steps_per_epoch = len(train_dataloader)
max_steps = total_epochs * steps_per_epoch
data_generator = iter(train_dataloader)
for global_step in range(max_steps):
    epoch = global_step // steps_per_epoch
    step = global_step % steps_per_epoch
    try:
        data = next(data_generator)
    except StopIteration:
        data_generator = iter(train_dataloader)
        data = next(data_generator)
    batch = rollout.rollout_batch(data, workflow=workflow)
    batch = batch.to(actor.device)

    logp = actor.compute_logp(batch)
    batch["prox_logp"] = logp
    if ref is not None:
        batch["ref_logp"] = ref.compute_logp(batch)
    actor.compute_advantages(batch)
    stats = actor.ppo_update(batch)
    actor.step_lr_scheduler()
    print(epoch, step, global_step, stats)
```

Abstraction Level 2: Algorithm Implementation

- How to train the model
- How to get and process results from forward passes
- Examples:
 - Advantage bootstrapping: RLOO, GAE, grouped normalization
 - Scoring logic of a PRM
 - Mini-batched update
 - PPO actor/critic loss
 - Algorithm-specific statistics
 - ...

Abstraction Level 2: Algorithm Implementation

Decouple backends with algorithm implementation

```
class PPOActor:

    def __init__(self, config: PPOActorConfig, engine: TrainEngine):
        self.config = config
        self.engine = engine

    @torch.no_grad()
    def compute_logp(
        self,
        data: TensorDict,
        temperature: Optional[float] = None,
    ) -> torch.Tensor | None:
        self.engine.eval()
        return self.engine.forward(input_=data, ...)

    def ppo_update(self, data: TensorDict):
        mb_inputs = split_padded_tensor_dict_into_mb_list(data)
        for mb in mb_inputs.mbs:
            train_stat = self.engine.train_batch(
                mb,
                loss_fn=...,
                ...
            )
            all_stats.append(train_stat)
        return all_stats
```

```
class FSDPPPOActor(FSDPEngine):

    def __init__(self, config: PPOActorConfig):
        super().__init__(config)
        self.actor = PPOActor(config, self)

    @torch.no_grad()
    def compute_logp(self, *args, **kwargs) -> torch.Tensor | None:
        return self.actor.compute_logp(*args, **kwargs)

    def ppo_update(self, *args, **kwargs) -> List[Dict[str, float]]:
        return self.actor.ppo_update(*args, **kwargs)

class MegatronPPOActor(MegatronEngine):

    def __init__(self, config: PPOActorConfig):
        super().__init__(config)
        self.actor = PPOActor(config, self)

    ...
```

Abstraction Level 3: Backends

- Providing unified interfaces for:
 - Training: FSDP, Megatron, DeepSpeed
 - Inference: SGLang/vLLM

```
class TrainEngine(abc.ABC):  
  
    def train_batch(  
        self,  
        input_: TensorDict,  
        loss_fn: Callable[[torch.Tensor, TensorDict], torch.Tensor],  
        loss_weight_fn: Callable[[TensorDict], float],  
    ) -> Dict[str, float]:  
        raise NotImplementedError()  
  
    @torch.no_grad()  
    def forward(  
        self,  
        input_: TensorDict,  
        output_seqlens: List[int] | None = None,  
        post_hook: Callable | None = None,  
        aggregate_fn: Callable[[List[Any]], Any] = torch.cat,  
    ) -> Any | None:  
        raise NotImplementedError()
```

```
class InferenceEngine(abc.ABC):  
  
    async def agenerate(self, req: ModelRequest) -> ModelResponse:  
        """Asynchronously generate a response."""  
        raise NotImplementedError()  
  
    def update_weights(self, meta: WeightUpdateMeta) -> Future:  
        """Update weights after training."""  
        raise NotImplementedError()  
  
    def submit(  
        self,  
        data: Dict[str, Any],  
        workflow: Optional["RolloutWorkflow"] = None,  
        workflow_builder: Optional[Callable] = None,  
    ) -> None:  
        """Asynchronously submit a request."""  
        raise NotImplementedError()  
  
    def wait(  
        self,  
        count: int,  
        timeout: float | None = None,  
        should_accept: Callable | None = None,  
    ) -> TensorDict:  
        """Wait for a specified number of requests to complete."""  
        raise NotImplementedError()
```

Agent Abstraction: the RolloutWorkflow class

- RolloutWorkflow
 - Inputs a (single, not batched) prompt
 - Outputs the trajectory used for training
- Easy workflow execution
 - The `submit` API
- Allows arbitrary agent implementation
 - Use `engine.agenerate` for completion
 - Arbitrary tool use inside the workflow

```
class InferenceEngine(abc.ABC):

    async def agenerate(self, req: ModelRequest) -> ModelResponse:
        """Asynchronously generate a response."""
        raise NotImplementedError()

    def submit(
        self,
        data: Dict[str, Any],
        workflow: RolloutWorkflow) -> None:
        """Asynchronously submit a request."""
        raise NotImplementedError()

    def wait(
        self,
        count: int,
        timeout: float | None = None,
        should_accept: Callable | None = None,
    ) -> TensorDict:
        """Wait for a specified number of requests to complete."""
        raise NotImplementedError()

class RolloutWorkflow:

    async def arun_episode(
        self, engine: "InferenceEngine", data: Dict[str, Any]
    ) -> Union[TensorDict, None]:
        raise NotImplementedError()
```

Agent Abstraction: the RolloutWorkflow class

```
def main_grpo():
    config, _ = load_expr_config(sys.argv[1:], GRPOConfig)
    config: GRPOConfig
    rank = int(os.getenv("RANK"))
    world_size = int(os.getenv("WORLD_SIZE"))
    tokenizer = load_hf_tokenizer(config.tokenizer_path)
    # Create dataset and dataloaders
    train_dataloader = StatefulDataLoader(
        get_gsm8k_dataset("train", rank, world_size),
        batch_size=config.train_dataset.batch_size // world_size,
        shuffle=config.train_dataset.shuffle,
        num_workers=config.train_dataset.num_workers,
        collate_fn=lambda x: x,
        drop_last=config.train_dataset.drop_last,
    )
    # Initialize inference engine
    rollout = RemoteSGLangEngine(config.rollout)
    # Initialize train engine
    actor = FSDPPPOActor(config=config.actor)
    ref = None
    if config.actor.kl_ctl > 0 and config.ref is not None:
        ref = FSDPPPOActor(config=config.ref)
    workflow = RLVRWorkflow(
        reward_fn=gsm8k_reward_fn,
        gconfig=config.gconfig,
        tokenizer=tokenizer,
        enable_thinking=False,
    )
```

```
# Run training.
total_epochs = config.total_train_epochs
steps_per_epoch = len(train_dataloader)
max_steps = total_epochs * steps_per_epoch
data_generator = iter(train_dataloader)
for global_step in range(max_steps):
    epoch = global_step // steps_per_epoch
    step = global_step % steps_per_epoch
    try:
        data = next(data_generator)
    except StopIteration:
        data_generator = iter(train_dataloader)
        data = next(data_generator)
    batch = rollout.rollout_batch(data, workflow=workflow)
    batch = batch.to(actor.device)

    logp = actor.compute_logp(batch)
    batch["prox_logp"] = logp
    if ref is not None:
        batch["ref_logp"] = ref.compute_logp(batch)
    actor.compute_advantages(batch)
    stats = actor.ppo_update(batch)
    actor.step_lr_scheduler()
    print(epoch, step, global_step, stats)
```

Agent Abstraction: the RolloutWorkflow class



- Code provided by the framework
 - Distributed launcher
 - Batched rollout execution
 - Dataflow
 - Backend integration

- User Code
 - Algorithm orchestration
 - Concrete agentic workflow

Agent Abstraction: the RolloutWorkflow class

AReaL



- *On the top of a single file, users can implement*
 - Any agent
 - Any orchestration code

While applications share the same AReaL train/inference engines for efficient asynchronous execution

AReal Examples: ASearcher

<https://github.com/inclusionAI/ASearcher/blob/main/AReal/ASearcher/train/asearcher.py>

```
class ASearcherWorkflow(RolloutWorkflow):

    async def collect_agent_trajectory(self, valid_inst, qid, prompt, prompt_token_ids, engine):
        agent = SearchAgent(prompt, prompt_token_ids)
        while agent.num_turns < self.max_turns and not agent.is_finished:
            # The agent prepares the prompt and sampling params for LLM generation
            input_ids, sampling_params = agent.prepare_llm_query(self.tokenizer)
            # Send request to inference engine and get response
            req = LLMRequest(...)
            if len(input_ids) + self.gconfig.max_new_tokens >= self.max_tokens:
                break
            resp = await engine.agenerate(req)
            # agent extracts tool callings from the llm response
            tool_calls = agent.consume_llm_response(resp, completion_str)
            # call tool and compute reward
            if tool_calls is not None and len(tool_calls) > 0:
                tool_call = tool_calls[0]
                res = (await self.toolbox.step((qid, [tool_call])))[0]

                agent.consume_tool_response(res, topk=self.topk)

        return ground_truth, score, agent.memory, stats
```

AReaL Examples: Generative Reward Models

```
class RLVRWorkflow(RolloutWorkflow):
    async def arun_episode(self, engine, data):
        text = self.tokenizer.apply_chat_template(
            data["messages"], tokenize=False, add_generation_prompt=True
        )
        req = LLMRequest(rid=uuid.uuid4().hex, text=text, gconfig=self.gconfig)
        resp = await engine.agenerate(req)
        rw_message = [{"role": "system", "content": "You should evaluate whether the response is correct or not."}]
        rw_text = self.tokenizer.apply_chat_template(
            rw_message, tokenize=False, add_generation_prompt=True
        )
        rw_req = LLMRequest(rid=uuid.uuid4().hex, text=rw_text, gconfig=self.gconfig)
        rw_resp = await engine.agenerate(rw_req)
        reward = parse_result(rw_resp.completions)
        ...
        return TensorDict(res, batch_size=[1])
```

AsyncGRPO with several lines of change

```
def main_grpo():
    config, _ = load_expr_config(sys.argv[1:], GRPOConfig)
    config: GRPOConfig
    rank = int(os.getenv("RANK"))
    world_size = int(os.getenv("WORLD_SIZE"))
    tokenizer = load_hf_tokenizer(config.tokenizer_path)
    # Create dataset and dataloaders
    train_dataloader = StatefulDataLoader(
        get_gsm8k_dataset("train", rank, world_size),
        batch_size=config.train_dataset.batch_size // world_size,
        shuffle=config.train_dataset.shuffle,
        num_workers=config.train_dataset.num_workers,
        collate_fn=lambda x: x,
        drop_last=config.train_dataset.drop_last,
    )
    # Initialize inference engine
    rollout = RemoteSGLangEngine(config.rollout)
    # Initialize train engine
    actor = FSDPPPOActor(config=config.actor)
    ref = None
    if config.actor.kl_ctl > 0 and config.ref is not None:
        ref = FSDPPPOActor(config=config.ref)
    workflow = RLVRWorkflow(
        reward_fn=gsm8k_reward_fn,
        gconfig=config.gconfig,
        tokenizer=tokenizer,
        enable_thinking=False,
    )
```

```
# Run training.
total_epochs = config.total_train_epochs
steps_per_epoch = len(train_dataloader)
max_steps = total_epochs * steps_per_epoch
data_generator = iter(train_dataloader)
for global_step in range(max_steps):
    epoch = global_step // steps_per_epoch
    step = global_step % steps_per_epoch
    try:
        data = next(data_generator)
    except StopIteration:
        data_generator = iter(train_dataloader)
        data = next(data_generator)
    batch = rollout.rollout_batch(data, workflow=workflow)
    batch = batch.to(actor.device)

    logp = actor.compute_logp(batch)
    batch["prox_logp"] = logp
    if ref is not None:
        batch["ref_logp"] = ref.compute_logp(batch)
    actor.compute_advantages(batch)
    stats = actor.ppo_update(batch)
    actor.step_lr_scheduler()
    print(epoch, step, global_step, stats)
```

AsyncGRPO with several lines of change

```
def main_grpo():
    config, _ = load_expr_config(sys.argv[1:], GRPOConfig)
    config: GRPOConfig
    rank = int(os.getenv("RANK"))
    world_size = int(os.getenv("WORLD_SIZE"))
    tokenizer = load_hf_tokenizer(config.tokenizer_path)
    # Create dataset and dataloaders
    train_dataloader = StatefulDataLoader(
        get_gsm8k_dataset("train", rank, world_size),
        batch_size=config.train_dataset.batch_size // world_size,
        shuffle=config.train_dataset.shuffle,
        num_workers=config.train_dataset.num_workers,
        collate_fn=lambda x: x,
        drop_last=config.train_dataset.drop_last,
    )
    # Initialize inference engine
    rollout = RemoteSGLangEngine(config.rollout)
    # Initialize train engine
    actor = FSDPPPOActor(config=config.actor)
    ref = None
    if config.actor.kl_ctl > 0 and config.ref is not None:
        ref = FSDPPPOActor(config=config.ref)
    workflow = RLVRWorkflow(
        reward_fn=gsm8k_reward_fn,
        gconfig=config.gconfig,
        tokenizer=tokenizer,
        enable_thinking=False,
    )
```

```
# Run training.
total_epochs = config.total_train_epochs
steps_per_epoch = len(train_dataloader)
max_steps = total_epochs * steps_per_epoch
data_generator = iter(train_dataloader)
for global_step in range(max_steps):
    epoch = global_step // steps_per_epoch
    step = global_step % steps_per_epoch
    batch = rollout.prepare_batch(
        train_dataloader,
        workflow=workflow,
    )
    batch = batch.to(actor.device)

    logp = actor.compute_logp(batch)
    batch["prox_logp"] = logp
    if ref is not None:
        batch["ref_logp"] = ref.compute_logp(batch)
    actor.compute_advantages(batch)
    stats = actor.ppo_update(batch)
    actor.step_lr_scheduler()
    print(epoch, step, global_step, stats)
```

AsyncGRPO with DAPO reward filtering

```
def main_grpo():
    config, _ = load_expr_config(sys.argv[1:], GRPOConfig)
    config: GRPOConfig
    rank = int(os.getenv("RANK"))
    world_size = int(os.getenv("WORLD_SIZE"))
    tokenizer = load_hf_tokenizer(config.tokenizer_path)
    # Create dataset and dataloaders
    train_dataloader = StatefulDataLoader(
        get_gsm8k_dataset("train", rank, world_size),
        batch_size=config.train_dataset.batch_size // world_size,
        shuffle=config.train_dataset.shuffle,
        num_workers=config.train_dataset.num_workers,
        collate_fn=lambda x: x,
        drop_last=config.train_dataset.drop_last,
    )
    # Initialize inference engine
    rollout = RemoteSGLangEngine(config.rollout)
    # Initialize train engine
    actor = FSDPPPOActor(config=config.actor)
    ref = None
    if config.actor.kl_ctl > 0 and config.ref is not None:
        ref = FSDPPPOActor(config=config.ref)
    workflow = RLVRWorkflow(
        reward_fn=gsm8k_reward_fn,
        gconfig=config.gconfig,
        tokenizer=tokenizer,
        enable_thinking=False,
    )
```

```
# Run training.
total_epochs = config.total_train_epochs
steps_per_epoch = len(train_dataloader)
max_steps = total_epochs * steps_per_epoch
data_generator = iter(train_dataloader)
for global_step in range(max_steps):
    epoch = global_step // steps_per_epoch
    step = global_step % steps_per_epoch
    batch = rollout.prepare_batch(
        train_dataloader,
        workflow=workflow,
        should_accept=lambda s: s["rewards"].mean() > 0,
    )
    batch = batch.to(actor.device)

    logp = actor.compute_logp(batch)
    batch["prox_logp"] = logp
    if ref is not None:
        batch["ref_logp"] = ref.compute_logp(batch)
    actor.compute_advantages(batch)
    stats = actor.ppo_update(batch)
    actor.step_lr_scheduler()
    print(epoch, step, global_step, stats)
```

Other Features

- Ecosystem-friendly
 - Agnostic to distributed scheduling backends



Other Features

- Pluggable
 - You can use AReaL outside AReaL
 - e.g., for data processing with asynchronous rollout

```
from areal.engine.sglang_remote import RemoteSGLangEngine
import os

dataloader = ...
engine = RemoteSGLangEngine()
os.environ['AREAL_SGLANG_SERVER_ADDRS'] = 'xxx,xxx'
engine.initialize(None, None)
results = engine.prepare_batch(dataloader)
```

- You can also import any packages with in the rollout workflow
 - E.g., rllm, math-verify, mcp, langgraph, ...

Other Features

- Quick debugging
- OpenAI-compatible rollout
- Automatic multi-turn discounting
- Automatic failover
- Dynamic sequence packing

***Contributions & collaborations
are welcomed!***

Q&A

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2025 Q3 Roadmap: <https://github.com/inclusionAI/AReaL/issues/257>

Paper link: <https://arxiv.org/pdf/2505.24298>

Code: <https://github.com/inclusionAI/AReaL>